

JUSTIN KIM

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justinkim.vercel.app

Education

University of Wisconsin, Madison | GPA: 3.76

Expected Graduation: Dec 2026

Bachelor of Science in Computer Science & Data Science

Madison, WI

- **Relevant Coursework:** Algorithms, Artificial Intelligence, Big Data Systems, Database Management Systems, Computer Networks, Machine Organization, Data Science Modeling I & II, Linear Algebra, Discrete Mathematics

Work Experience

Capital One

June 2026 – Aug 2026

Software Engineer Intern

Chicago, IL

- Architected a graph-RAG pipeline that translated plain-English requirements into validated fulfillment workflow configs, reducing build time from 2+ days to under 1 hour for intent engineers.

MadLease

Jan 2024 – Present

Co-Founder & Founding Engineer

Madison, WI

- Founded and deployed a UW–Madison sublease marketplace with 200+ accounts, 90+ listings, and 5+ building partners.
- Improved sublease discovery by designing a ranking engine that matched listings to users across 7 filter dimensions including price, location, amenities, and date availability.
- Boosted listing visibility across 25+ crawler types by serving structured metadata via Vercel Edge Middleware, while routing real users directly to the React SPA.
- Eliminated backend image traffic for 1000+ uploads by architecting browser-direct S3 transfers with presigned PUT URLs, CloudFront CDN delivery, and Firebase UID/IP token-bucket rate limiting.

Integral

May 2025 – Aug 2025

Full Stack Software Engineer Intern

Madison, WI

- Led development of a production-ready crash photo capture and insurance intake platform, creating a company-wide reusable OCR module that extracted VIN, license plate, and odometer fields with 90%+ accuracy.
- Eliminated polling for estimate negotiation data by building a Pub/Sub database listener that propagated updates across multiple internal services in real time.

Denari

Feb 2025 – May 2025

Software Engineer, Client Project

Madison, WI

- Built a citation-grounded RAG pipeline for tax document search, improving answer accuracy by 8% over a GPT-4o-mini baseline across 2,200+ CPA exam benchmark questions using OpenAI embeddings, TimescaleDB/pgvector, and AWS S3.

Astro AI Trading

Jan 2024 – Dec 2024

Machine Learning Engineer

Remote

- Led a 3-engineer team at an early-stage fintech startup building sentiment-augmented stock forecasting models trained on market data and 60+ daily news/social sources, reducing held-out RMSE by 32% over a price-only baseline.
- Built a BERT + PCA + KMeans article routing pipeline that clustered financial news by sector, improving sentiment signal quality for downstream forecasting models.

Projects

QuizClash | *Devpost* | *Source Code*

NodeJS | **React** | **WebSocket** | **AWS**

- Developed a real-time multiplayer quiz app that converts PDFs and YouTube videos into Kahoot-style games, generating playable multiple-choice sets from uploaded content with 75% manually evaluated relevance.
- Automated quiz generation with a 5-stage ML pipeline using OCR, Whisper, T5, RoBERTa, and Sense2Vec for content ingestion, question generation, answer extraction, and distractor selection.
- Powered real-time gameplay with a WebSocket engine supporting synchronized timers, answer locking, score reveals, and public/private room management.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C, SQL, HTML5, CSS

Developer Tools: Git, Docker, AWS (S3, EC2, IAM), Postman, Jira, WireMock, Pub/Sub

Libraries/Frameworks: React, Spring Boot, Node/Express, PostgreSQL, Flask, PyTorch, Angular, Scikit-Learn

Concepts: Machine Learning, Full Stack Development, Backend, Frontend, Data Science, Agile Methodologies